

CANDIDATE NAME: _____

INDEX NUMBER _____



SERANGOON JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION 2017

BIOLOGY 9744
Higher 2

CG _____

PAPER 3

**Friday
15 September 2017**

2 hours

Additional materials:
Optical Mark Sheet

READ THESE INSTRUCTIONS FIRST

Write your name and CG in the spaces at the top of this page.
Write in dark blue or black pen.
Do not use staples, paper clips, glue or correction fluid.
Do NOT write in any barcodes.

Section A

Answer all the questions in the spaces provided in the question paper.

Section B

Answer any **one** question in the spaces provided in the question paper.

FOR EXAMINER'S USE	
Paper 3	
1	/29
2	/21
3 or 4	/25
P3 Total	/75

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [] at the end of each question or part question.

Section A

Answer all the questions in this section.

Question 1

Repeat droughts may cause permanent damage to forests

By: [Alex Whiting](#)

ROME, Aug 9 (Thomson Reuters Foundation) - Trees and their environment need as long as two years to recover from drought in some places, and if a second dry spell hits before then, it may cause permanent damage to the landscape, researchers said on Wednesday.

With climate change expected to bring more frequent and intense droughts, the implications for areas that do not have time to bounce back fully could be severe, the researchers said in a paper to be published in *Nature* journal this week. "That could have a double whammy effect," said co-author William Anderegg, assistant professor of biology at the University of Utah. "A second drought could be harder on an ecosystem and have the potential to push it off a cliff."

In practice, that means affected areas could eventually turn from lush forest to a land of grass and shrubs. Boreal forests in northern parts of Europe, Russia and Canada can take up to two years to recover from drought, partly because they do not have a wide variety of plants, Anderegg told the Thomson Reuters Foundation. Forests in the tropics of South America and Southeast Asia have also taken the same amount of time to rebound.

"That's worrisome because those regions store the largest chunks of carbon in ecosystems across the globe," Anderegg said. Forests help tackle climate change by sucking carbon out of the air, reducing levels of planet-warming carbon dioxide, the main greenhouse gas. But when trees die, most of the carbon they have absorbed is released back into the atmosphere.

The Amazon rainforest suffered a double drought in the first decade of this century when dry spells, both of a once-in-a-100-years severity, hit the region. "Satellites showed that forests hadn't recovered from the 2005 drought by the time the 2010 drought struck," Anderegg said.

Adapted from a Reuters Article

- (a) Discuss how anthropogenic climate change has affected the growth of forests in the world. [4]

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- (b) The article talks about how forests may not be able to recover if a second drought soon after a first. Suggest how forests may recover from a drought and explain why a second drought soon after a first impedes its ability to recover. [4]

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- (c) The article talks about forests storing carbon as shown by the following quote "*That's worrisome because those regions store the largest chunks of carbon in ecosystems across the globe,*" Anderegg said. Define the role of forests in storing carbon and explain briefly how they are able to do this. [4]

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Climate change does not just affect our forests, it also affects our farms. The following article elaborates on this.

Aug. 4 (UPI) -- Scientists from Lancaster University suggest major changes in agricultural practices are needed to offset increases in nutrient losses due to climate change. The study, shows that phosphorus losses will continue to increase due to climate change unless major changes in agricultural practices are made.

"Although farmers are already doing what they can to prevent these losses, the currently adopted measures are not likely to be enough to offset the increase expected under climate change. This paper should alert policy makers and government to the help and support that farmers will need to achieve the scale of agricultural change that may be necessary to keep up with the increase in pollution due to climate change."

Although phosphorus and nitrogen are essential to crop and animal growth, too much of it can cause algae blooms in rivers and lakes.

- (d) Phosphorus and nitrogen are essential nutrients for plant growth and are typically found in large concentrations within the fertilizers used on farms. Suggest how climate change can lead to phosphorus and nitrogen from fertilizers ending up in rivers and water bodies. [2]

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- (e) The article talks about nitrogen and phosphorus causing algal blooms in rivers. Using a specific example of a plant or animal, discuss the possible impact of such blooms on the natural ecosystem in these rivers. [4]

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- (f) The article suggests that farmers need to change their agricultural practices to limit the impact of phosphorus used in fertilizer on nearby aquatic bodies. Describe two changes they could make and suggest why it may be difficult for them to make such changes. [3]

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Another aquatic ecosystem affected by climate change are coral reefs. The following graph, **Figure 1**, shows how sea water temperatures have varied over the last century or so at the Great Barrier Reef. 0°C is considered the average sea water temperature.

Rising sea level temperatures have been suggested as a reason for more frequent coral bleaching events. **Figure 2** shows the trend for coral bleaching events globally since 1980.

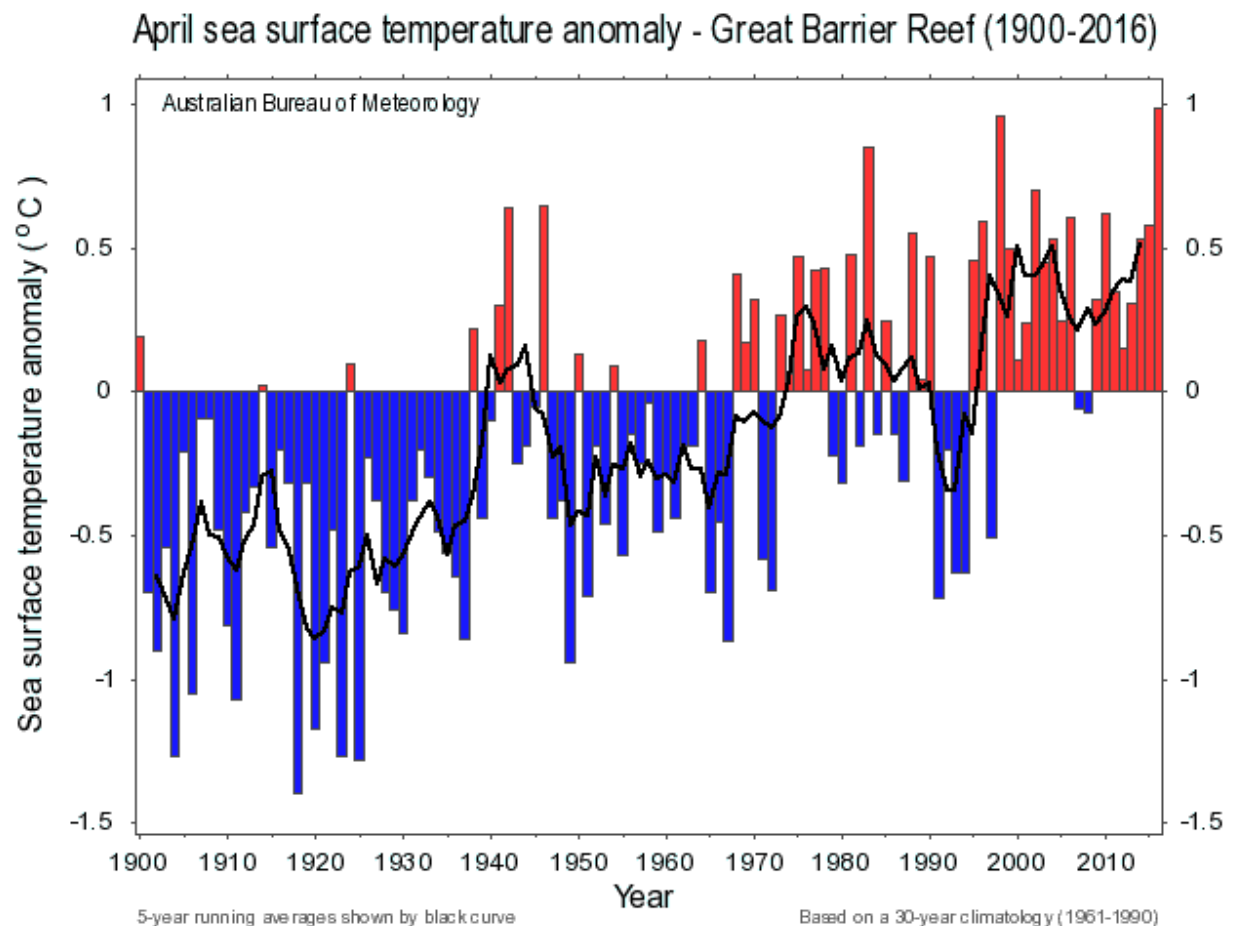


Figure 1

(h) Other than warming water temperatures, describe two other ways anthropogenic climate change has impacted coral reefs. [2]

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[Total: 29 marks]

Question 2

A method that has been used to sequence DNA is called Sanger Sequencing, named after the scientist that invented the process. Sequencing DNA essentially allows the sequence of bases on a DNA strand to be read. During Sanger sequencing, DNA polymerases copy single-stranded DNA templates by adding nucleotides to a growing chain (**extension product**). Chain elongation occurs at the 3' end of a primer, an oligonucleotide that anneals to the template. The extension product grows by the formation of a phosphodiester bridge between the 3'-hydroxyl group on the primer and the 5'-phosphate group of the incoming deoxynucleotide.

DNA polymerases can also incorporate analogues of nucleotide bases. The dideoxy method of DNA sequencing developed by Sanger et al. 1977 takes advantage of this characteristic by using 2',3'-dideoxynucleotides (ddNTPs) as substrates. When dideoxynucleotides are incorporated at the 3' end of the growing chain, chain elongation is terminated selectively at A, C, G, or T. This leads to the production of a DNA fragment with the ddNTP at its end.

The ddNTPs are usually labelled using fluorescent tags. Many fragments of the original DNA strand terminated in the manner shown in **Figure 3** are created using this processes. When all these fragments are separated on a gel using a form of gel electrophoresis, the sequence of bases can be read by reading the labelled ddNTPs from the shortest to the largest fragment.

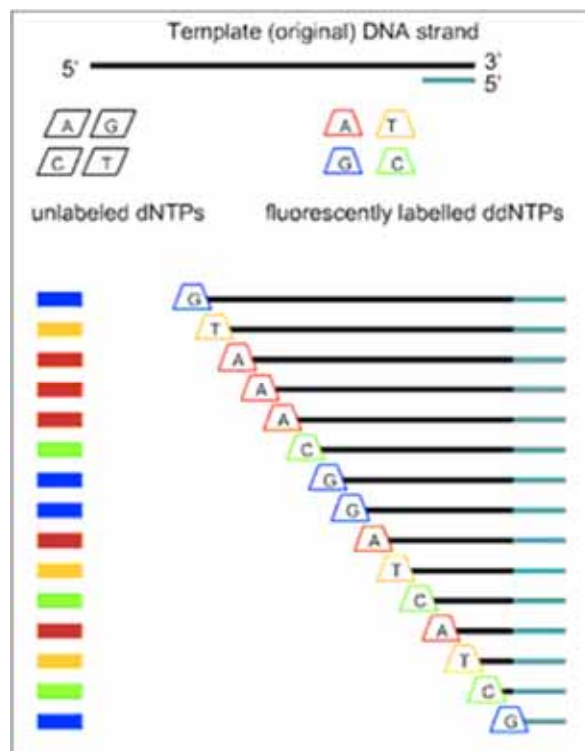


Figure 3

(a) Suggest how the labelling of the ddNTPs allows for the identification of the base that normally occupies that position. [2]

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(b) Describe how the extension product (line 4) is created and explain the role of the template DNA strand to this process. [5]

[illegible]

(c) Explain how, when separated using gel electrophoresis, the strands produced by Sanger Sequencing end up in different positions on the gel. [3]

[illegible]

The following **Figure 4** shows an example of dideoxynucleotide (ddNTP).

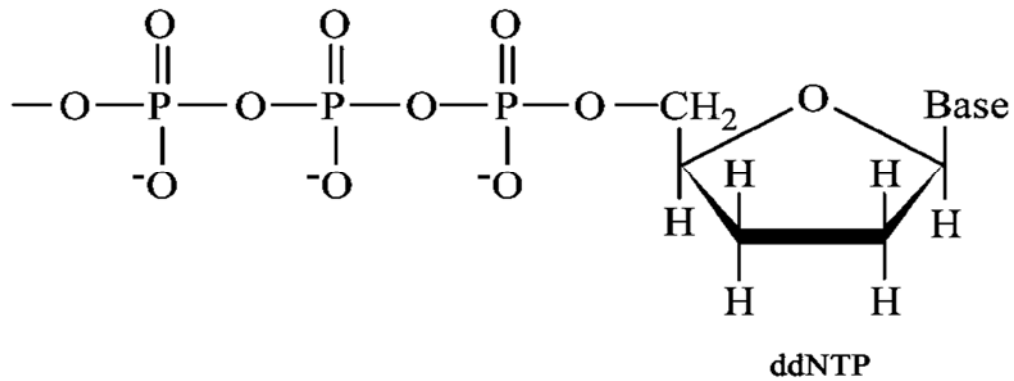


Figure 4

(d) Explain why dideoxynucleotides, such as the one shown in Figure 4, leads to the formation of fragmented DNA. [3]

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- This image shows a single sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

(f) Suggest two ways in which mitochondrial DNA may differ from the nuclear genome. [2]

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(g) With reference to specific genes, explain the role of mitochondrial DNA. [2]

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[Total: 21 marks]

Section B (25 Marks)

Answer **only one** question

Write your answers in the space provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose where appropriate.

Your answers must be set out in sections (a), (b), etc as indicated in the question.

Question 3

- (a) All organisms need to replicate and transcribe/translate their DNA in the process of growth and development. Even viruses need to do so. Using the example of a dengue virus, compare and contrast the replication and protein synthesis process in a virus with the similar processes in a typical eukaryotic cell. [13]
- (b) Describe how viruses such as the influenza virus is able to create genetic variation and explain how this makes it difficult for us to eradicate harmful viruses with modern medicine. [12]

Question 4

- (a) All organisms need to control the expression of their DNA in the process of growth and development. Even bacteria need to do so. Using the example of *E. coli*, compare and contrast the control of gene expression in a bacteria with the similar processes in a typical eukaryotic cell. [13]
- (b) Describe how bacterial cells are able to create genetic variation and explain how this makes it difficult for us to eradicate harmful bacteria with modern medicine. [12]

End of Paper